

IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

EXAMINATIONS 2025

BEng Honours Degree in Computing Part II
MEng Honours Degrees in Computing Part II
BEng Honours Degree in Mathematics and Computer Science Part II
MEng Honours Degree in Mathematics and Computer Science Part II
for Internal Students of the Imperial College of Science, Technology and Medicine

*This paper is also taken for the relevant examinations for the
Associateship of the City and Guilds of London Institute*

PAPER COMP50004

OPERATING SYSTEMS

Wednesday 7th May 2025, 10:00

Duration: 90 minutes

Answer ALL TWO questions

Paper contains 2 questions
Calculators required

- 1 You are asked by your manager to implement a *scheduler* component for a new experimental operating system (OS) that runs on smart home devices.

A smart home device has 4 CPU cores and various I/O hardware. A device must typically support a total of 1000 concurrent processes, which can all interact with the I/O hardware. About 10 processes control interactive features that are part of the device's user interface and thus require a low response time for a good user experience.

(Regarding any information not given in the questions, you should make reasonable assumptions and state them as part of your answer.)

- a State **four** distinct goals that a general OS scheduler should achieve, briefly justifying each.
 - b Would you implement a *preemptive* or *non-preemptive* scheduler for the above scenario? Briefly justify your answer.
 - c Explain what *scheduling algorithm* you would use for the above scenario and give reasons why it is a suitable choice.
 - d Describe the *data structures* that you would use in your scheduler implementation, and justify your choices. When selecting data structures, you should comment on their time complexity in Big-O notation.
 - e Write down your *scheduler implementation* in C-like pseudocode.
- Your implementation should use the above data structures and provide code for three cases: (i) a timer interrupt triggers; (ii) a process invokes an I/O operation; and (iii) the I/O operation invoked by a process has completed.

(You can assume the existence of relevant helper functions, and you will not lose marks for minor syntactic mistakes.)

The five parts carry, respectively, 20%, 10%, 15%, 20%, and 35% of the marks.

- 2a We have a memory translation subsystem that uses paging to translate virtual addresses into physical addresses, where each entry on any level of the page table occupies 8 Bytes. The TLB has 8 entries for 4 KB pages, and takes 10 nsec for every lookup. The CPU has no instruction or data caches, and every main memory access takes 500 nsec.
- i) Calculate the average memory access time for a 1-level page table when we know that the TLB hit rate is 90%.
 - ii) Calculate the average memory access time for a 4-level page table when we know that the TLB hit rate is 90%.
 - iii) Identify and briefly explain if you would use the 1-level or 4-level page table on each of the following cases:
 - 1) 32-bit virtual addresses, 20-bit physical addresses, 1 GB worth of data mapped and actively used by applications (swapped-in).
 - 2) 64-bit virtual addresses, 20-bit physical addresses, 1 GB worth of data mapped and actively used by applications (swapped-in).
- b We have a disk formatted with an *ext2* file system that we use to store a 50 GB file. The file system is configured with 4096 Byte blocks, 8-Byte indirection pointers, and a root file inode with 12 direct pointers, and 3 pointers to the corresponding indirect blocks (from singly to triply indirect block).
- Calculate the maximum number of disk block access operations we will need in the worst case to read an arbitrary position in the file. Show your working, assuming our first access is always to the inode structure.
- c Name and briefly describe one advantage and one disadvantage for each of the following forms of machine virtualization:
- 1) binary translation
 - 2) paravirtualization
 - 3) hardware support for machine virtualization

The three parts carry, respectively, 40%, 30%, and 30% of the marks.